Manual

Collection / Information Functions for Material, ERP Batches, MES Batches

AIP-MTR 8.1

Version 1.0.23049

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Contents

1 Overview of Collection / Info Functions for Material, ERP Batches and MES Batches

Purpose

The AIP features contained in this function package make it possible to enter batch-related data directly in production using shop floor terminals or data collection PCs.

Integration

The data entered using the AIP can be displayed in various applications or evaluated in the MOC. The data entered can also be uploaded via interfaces.

Features

Order-related data entry and posting functions

- Batches can be entered at the same time as operations can be posted
- Entry and validation check of input and output batch changes while the operation is running
- Entry of goods receipt batches and automatic generation of goods receipts/ goods issues
- Batch numbers can be entered at the same time that operations are logged on (configurable per operation and workplace)
- A ticket/ label is automatically printed in HYDRA standard format using the assigned printer when a new batch is generated
- A validation check for documentation requirement is run for batches when the operation is logged on
- Entry of batch changes during order processing
- Entry of batch numbers via keyboard and/ or barcode.
- Batch-related quantity and time input

Functions to enter/generate and display series and serial numbers:

- Predefined series or serial numbers are transferred in HYDRA standard format via the HYDRA ERP interface
- Serial numbers are entered or generated via a user-friendly posting function at the BDE terminal
- Possibility to assign components also identified by serial numbers
- Classification into yield/ scrap quantities with reasons
- Automatic generation of goods receipts/ goods issues
- Optional verification of serial numbers already defined for the order

- Entry of order-related series and serial numbers identifying the processed parts
- Validation check for already posted series or serial numbers
- Display of available and used series or serial numbers at workstation PCs (MES Operation Center)
- Upload of posted series and serial numbers in HYDRA standard format via the HYDRA-ERP interface

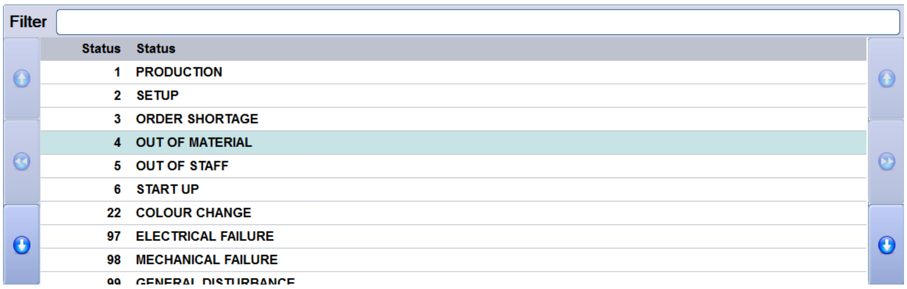
Additional licenses may be needed in order to use the functions listed above. Adding and coordinating the specific requirements and implementing them are considered a customized HYDRA service.

2 Operation of AIP

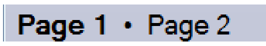
2.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, …) as well as the selection lists of posting dialogs.


Filter

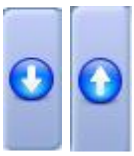
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE



Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys:'↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.


Scrolling page by page (up or down) in the table.


Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

A “table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.

The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+“, "-“ , ".“ and ",“. Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter –t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-“, “/“ and “.“ are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- English(USA) MM/DD/YYYY
- Danish MM-DD-YYYY
- Customer-specific 1 YYYY-MM-DD
- Customer-specific 2 YYYY/MM/DD
- Customer-specific 3 MM/YYYY/DD

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

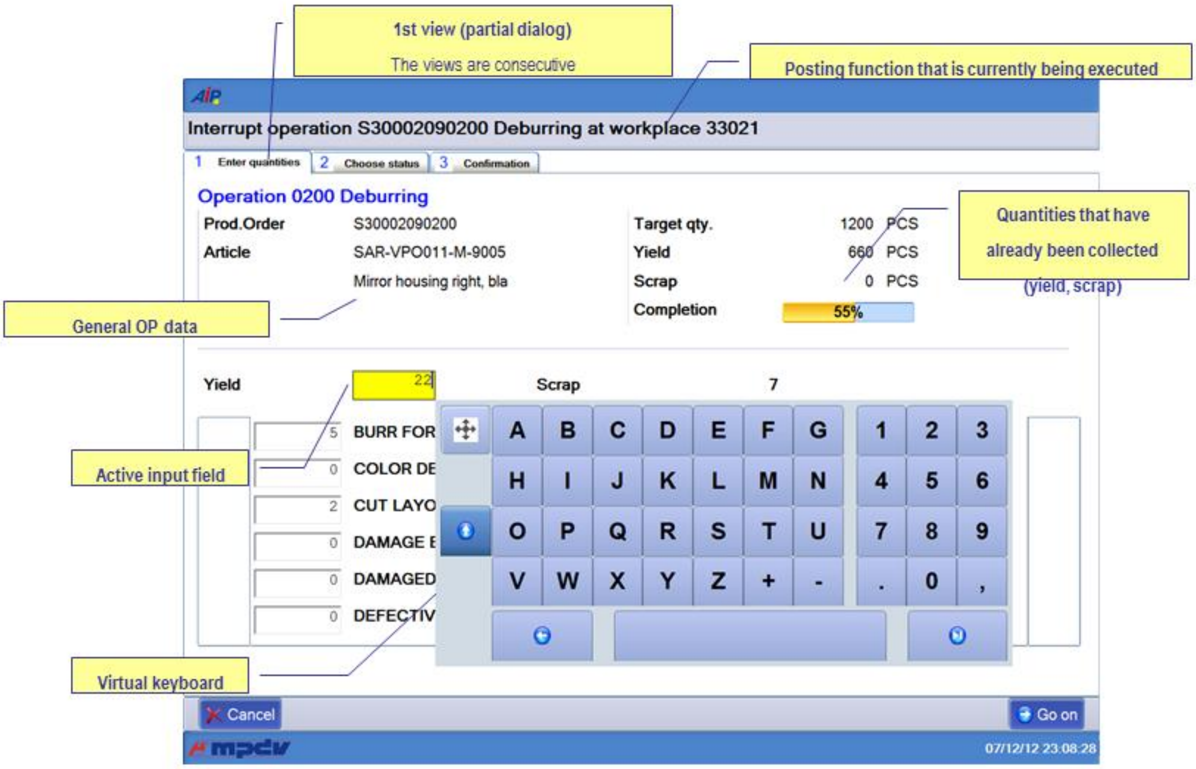
The year is displayed only by two characters in the status bar.

2.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:

The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.



The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

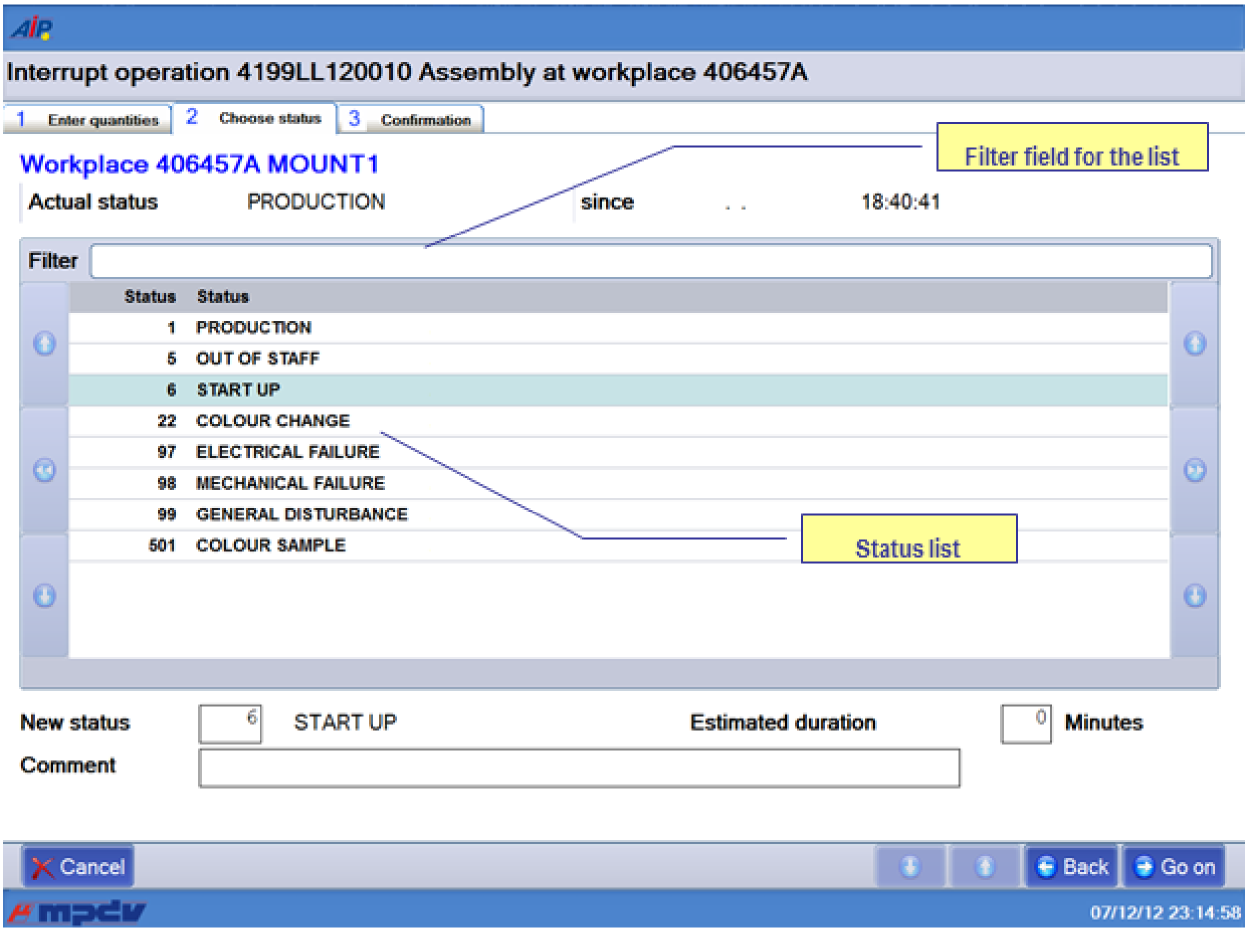
Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

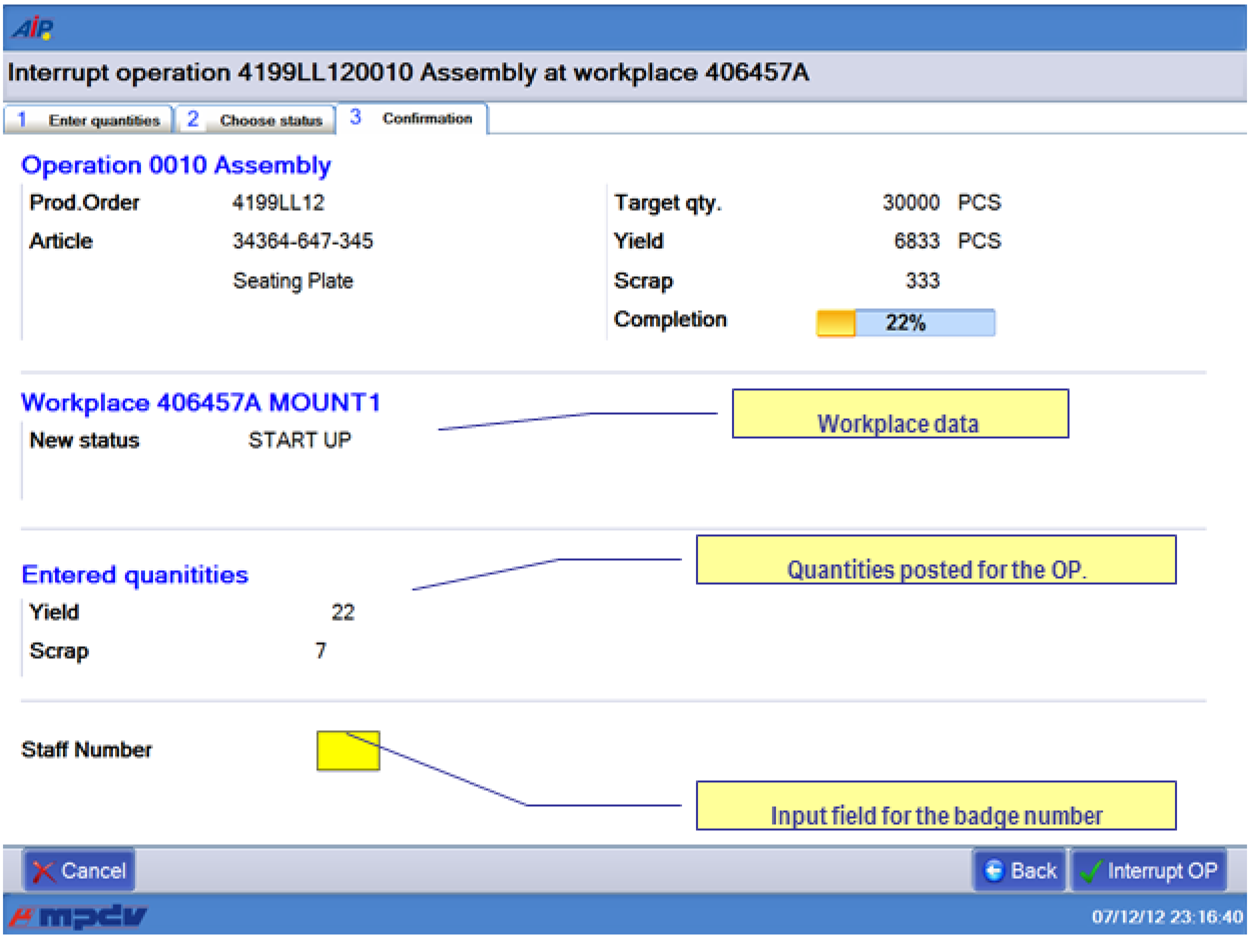
The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.



The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).



The partial dialog “confirm” shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the “interrupt operation” dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.

If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.

3 Basic AIP Display

	→	

In general, *AIP* has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons within the touch screen or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these selection lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.

To ensure proper processing and posting, terminals with "MDE" operation mode must <u>not</u> be switched off during times without shift.

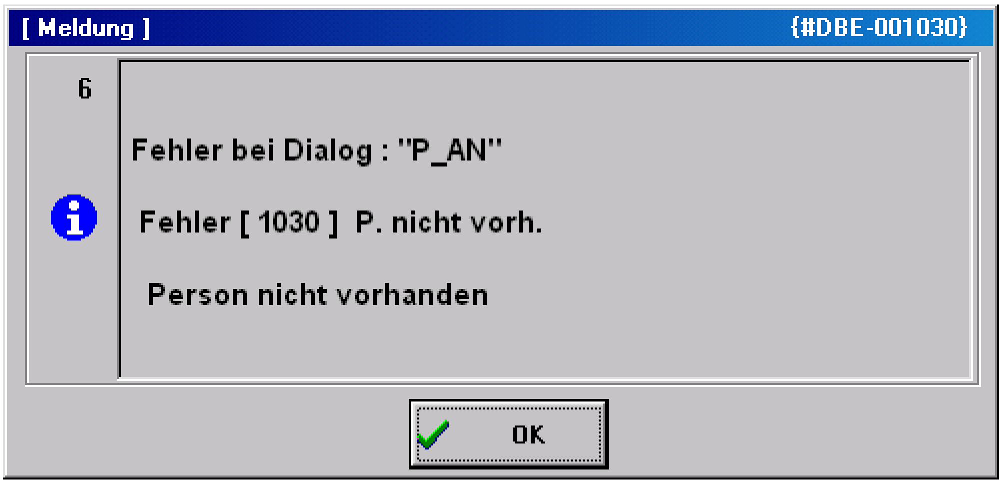
3.1 Basic displays – header and footer

Header

The *AIP* logo is displayed top left of the screen, which may be exchanged by a customer logo after corresponding configuration.

Possible messages (e.g. if a dialog is opened for more than five minutes) are displayed to the right of it.

A separate window opens to display error messages that occur during data collection (e.g. validity checks).



Basic displays

A maximum of 16 workplaces or machines can be assigned to the *AIP* terminal. The individual workplaces can be found within the list area in the order in which they have been assigned to the terminal in MOC.

As regards the basic display of the *AIP* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured within the terminal configuration at the client. The individual basic views are described in the sections that follow.

Footer

The MPDV logo can be found at the bottom left of the screen. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP version number, date of the build, IP address of the server as well as the terminal number.

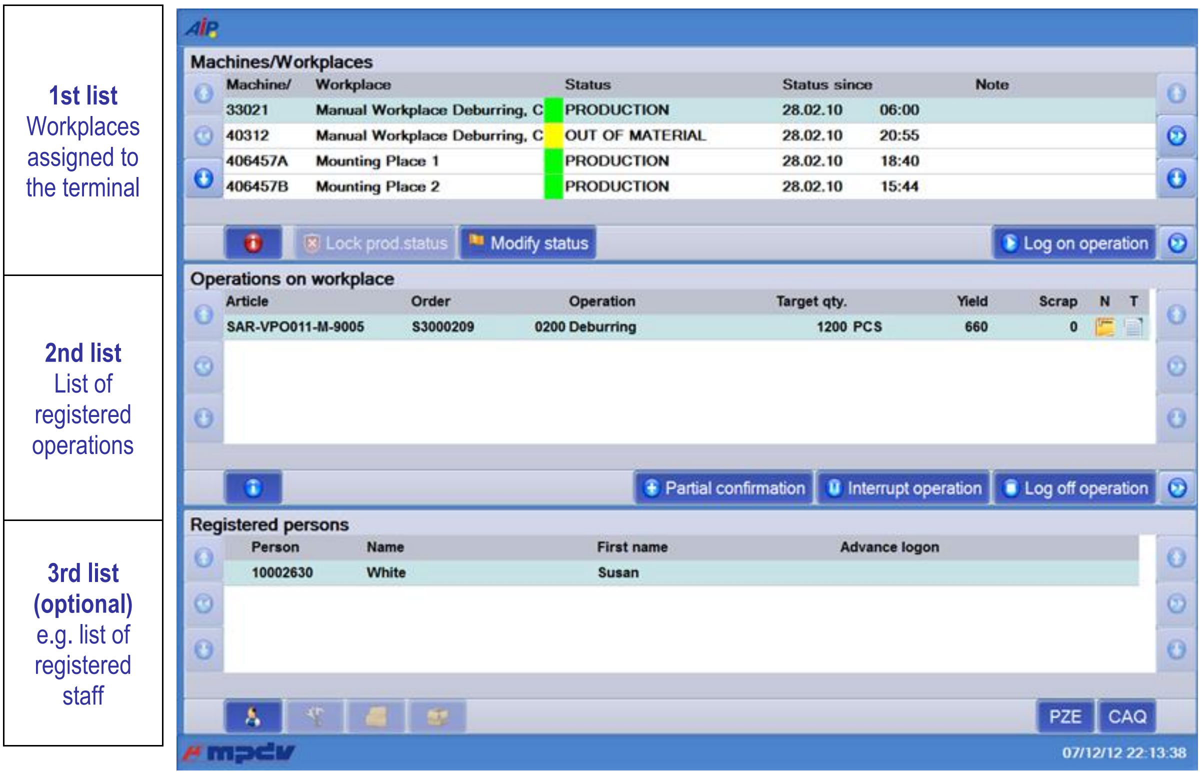
The current date and time are displayed to the right.

AIP status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	Commands are being transferred to the server
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.

	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

3.2 Basic display “tabular view“



The tabular basic display consists of two or three tables, subject to the configurations made. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional display).

“Machines/workplaces" table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as the designation are displayed.

Status

The status is displayed in color and as status text. Coloring is as follows:

- green: Production
- yellow: Assigned status
- red: Not assigned

Status since

Point in time since the status is available.

For ADE workplaces[1] the point in time refers to the last manual status change, for MDE workplaces it refers to the time when the last status change was identified (for machine connections), to the point in time when the status was changed manually the last time or to the time of the last shift change.

Please note

It is indicated here if the “block production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.

Please note

By way of “customizing” services it is possible to adapt the layout of display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic display of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately, once they have been entered but only once lists have been reloaded.

"Operations at workplace" table

The second table shows the operations that are currently logged on to the selected workplace. The following columns are displayed:

[1] We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an ADE workplace.

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number.*

Target quantity

Target quantity that is defined for the operation.

Yield

Yield which has already been produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity which has already been produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note, which should be visible at the terminal, has been recorded for this operation in the graphic planning board at the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a line that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured respectively. Which information is displayed in this list depends, among other things, on the workplace configuration.

The following lists can be displayed:

- Staff logged on to the currently selected workplace (BDE)
- Resources logged on to the currently selected workplace (WRM)

- Materials/input batches logged on to the currently selected workplace (MPL/TRT)

- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow for switching between these lists.

Please note

The registered staff displayed in the third list correspond to the list of the dialog “F5 registered persons…”. Selecting a person in the third list does *not* affect selection of the operation in the list of OPs running at the workplace and, as a result, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic display

A toolbar, which may be configured by customizing, is assigned to each list in the basic display. This makes the purpose of the function clear to the user. The “partial upload/confirmation” function can be found, for example, below the list of registered operations.

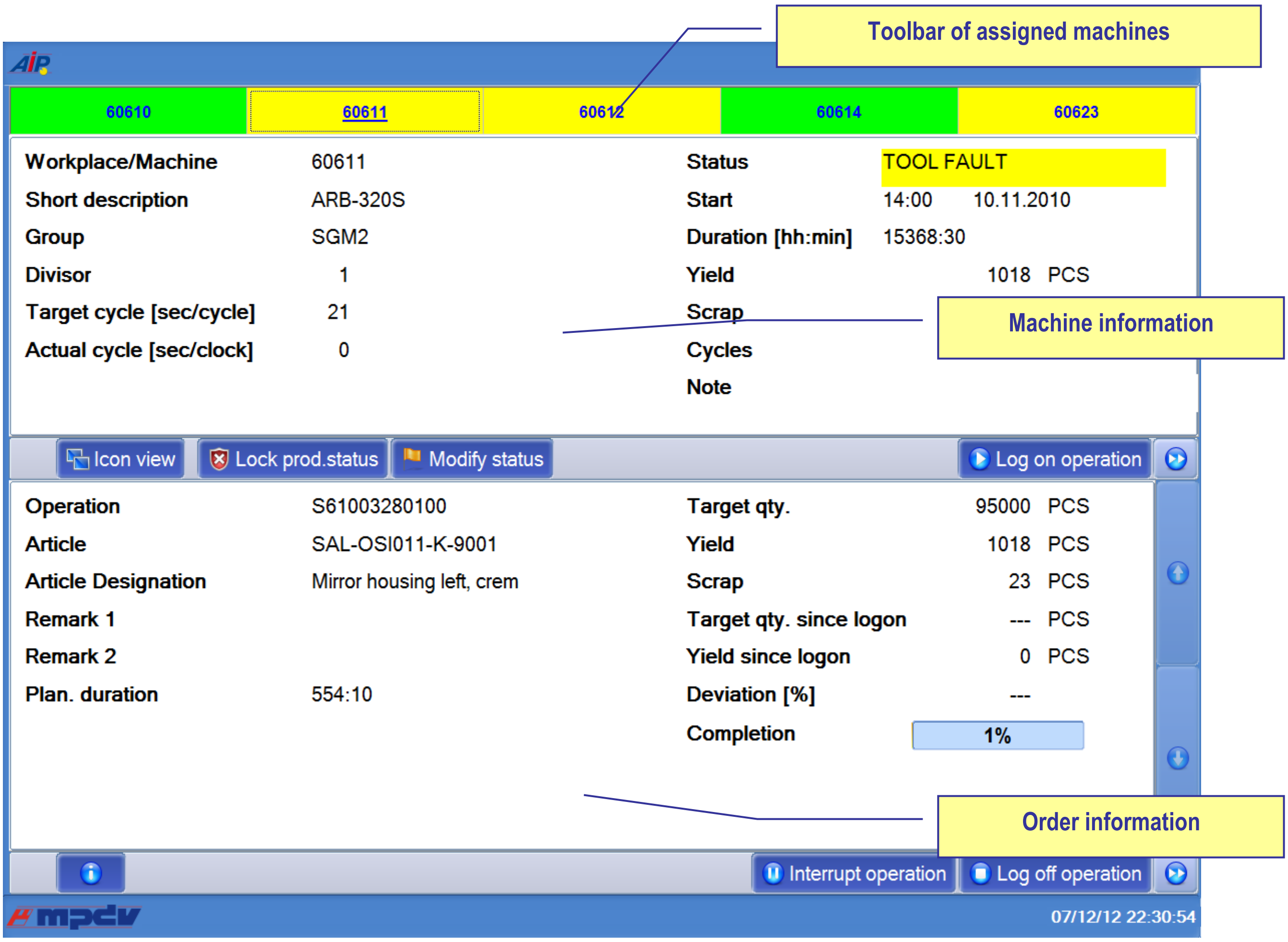
In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. change partitioning) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

3.3 "Machine overview" basic display

If the “change view” button is clicked in the basic display, the view changes to the following presentation:



This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation altogether has three areas:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines as well as to shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. Provided that several orders/OPs are logged on, it can be switched between them using arrow keys that are then shown additionally.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$Displayed\ partitioning = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \ldots \right]$$

TLG_M	Partitioning of the machine (TLG in mnr.lst)
DIV_M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG_{AGi}	Partitioning of the individual order (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview at the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal may get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) at the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time in which the production lock of the machine has not been active). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.
Calculation:
TargetSinceLogon = NetRunningTime[sec] * Partitioning/TargetCycle[sec/stroke]
NetRunningTime: Time since logon, in which the production lock has not been set. This calculation does not take into account any breaks that might be defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

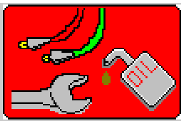
Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (YieldSinceLogon - TargetSinceLogon) / TargetSinceLogon

Completion

The bar represents the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced by a picture showing a yellow or red oilcan, depending on the maintenance activity that is required:



3.4 “Machines as icons” basic display

This view can be configured as the default display within the terminal configuration at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

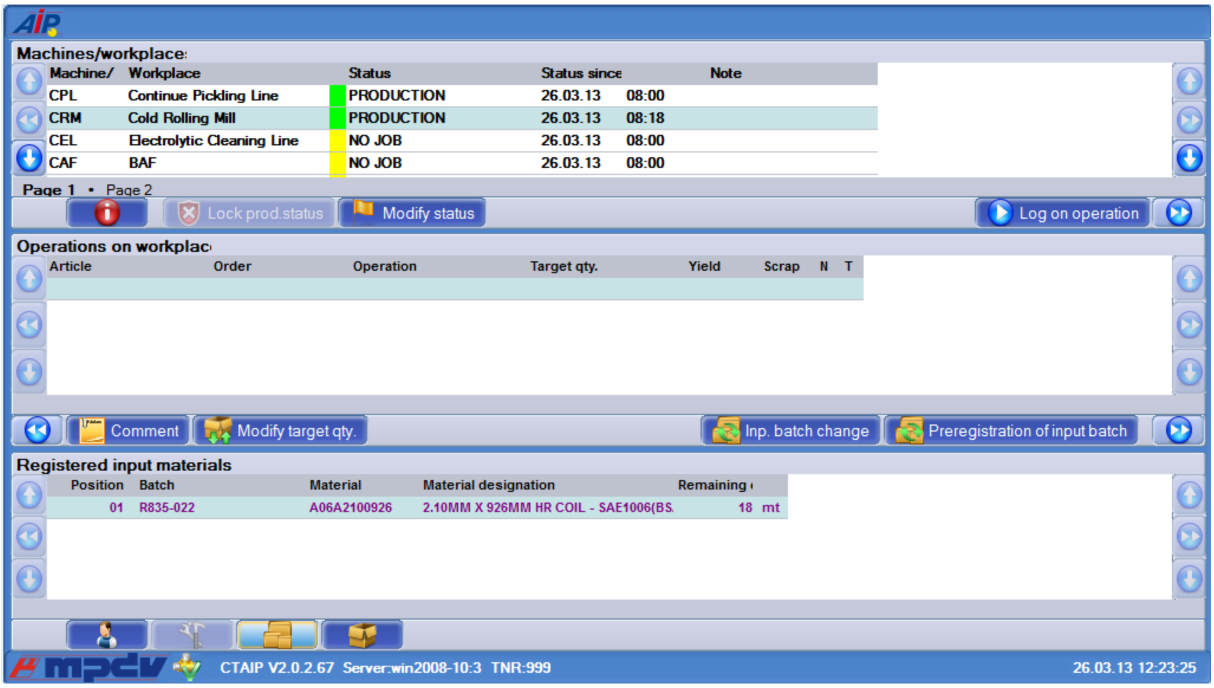
If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon presentation of machines”.

4 Input Functions Relating to Batches

4.1 Basic screen

Basic terminal screen when a machine is assigned in batch mode:



The basic display shows the third list "Input materials currently logged on" for machines for which the “batch management” option is configured. This list shows all active input batches of the selected machine.

4.2 Order postings for operations subject to batch management

OP logon with input batches

By clicking the "log OP on" button a workflow including two tabs is opened. The operation to be logged on is selected in the first tab “select operation”.

The “log operation on” tab is reached by clicking the “Next” button where in addition to the selected OP, the defined material components are displayed in a list.

By entering a batch number in the "input batch" field and clicking the "report batch" function a batch may be logged on as input material for a component. During the entry process, the terminal checks whether or not the batch number is known in the system and may be logged on. This is described in detail within the “input batch change" section.

“Batch” field

When an OP is logged on, a batch number is created simultaneously for the next output batch to be produced. The batch number may be assigned automatically or manually (please also see machine master settings --> MPL tab). The generated batch is created with the batch number in the system and set to the "running" status.

Provided that all required input materials are logged on, the OP may be started via the "OK" button in the "OP logon" dialog. Whether input material has to be logged on or not, can be defined in the assigned material type of the component.

Once the OP has been logged on successfully, all active input batches of the selected machine are displayed in the material list.

As of HYDRA-MPL product version 7.2.5:

If batches are logged on along with the OP and the user cancels the process or cannot log the OP on due to a plausibility check the input batches will be logged off automatically for this OP. In this case, batches are always logged off without indicating the consumption quantity.

By way of the following warning message, the user may confirm the logoff process:

The function that logs input batches off automatically can be activated/deactivated by a button in the hytnrcfg.ini file:

HYTNRCFG.INI
[MPL-Options 0/2xxx] ForceAutoLogOffInputBatches=0

Please note: This is only available as of HYDRA-MPL product version 7.2.5 and CTAIP version 2.0.2.3.

Logoff/interruption of OPs

A running OP may be interrupted or logged off by clicking the "logoff/interrupt OP" button. Then a dialog opens, where the following selection can be made:

If "log OP off" is clicked the logoff dialog opens that contains the same input fields like the "output batch change" dialog (see next section).

Thus, the output batch that is currently still active is completed, when OPs are interrupted or logged off.

4.3 Postings based on batches

Input batch change

Input material can be changed for a running OP if the "input batch change" button is clicked.

Log input batch off

Input batches may be changed by entering a currently active batch number or by entering a new batch number. When logging batches off, it is also possible to enter the status and consumption of the batch to be logged off.

Options when logging input batches off:

F1 - PROCESSED

The batch is set to the "processed" status and the remaining quantity that is still available is set to 0. A consumption posting is generated as goods issue for the current, remaining quantity.

F2 - BLOCKED

The batch is set to the "blocked" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue.

F3 - with remaining quantity

The batch is set to the "free" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue. If the remaining quantity that is still available becomes <= 0, the batch status automatically switches to "processed".

Consumption

The entered consumption (unit of the input material) is deducted from the remainder of the batch and a goods movement is generated.

Comment on batch

The entered comment is saved as information for the batch.

Log input batch on

Provided that the batch is known, batch data are displayed in an intermediate dialog where the logon may be confirmed.

Provided that the batch could be logged on, it is taken over to the material list in "customer batch number" and thus the change is completed.

However, in case the logon is inadmissible as the input material does not correspond to that of the component, the logon is rejected by the following error message:

Logon of unplanned input material

In addition to planned materials, it is also possible to log “unplanned” material on for an OP, using an additional feature at the component of the OP. If the "replaceable" option is set to "J" the user is able to assign the respective component manually, when batches are logged on. However, the logon is only allowed if the material type of the input batch corresponds to that of the component.

Within the selection list the components are filtered according to the material number and displayed as follows:

Logon of unknown batches

An input batch, which is not yet known in HYDRA, may be logged on for an OP using the "creating unknown batches" option in the basic settings of HYDRA (in BDE --> settings 2 tab):

In this case, it is searched for a valid assignment of input material to the material type of the selected component, when input batches are logged on. Provided that a corresponding assignment is found and the "allow entry of unknown input batches" option is configured in the "input batch processing" tab at the material type, the batch is generated by logging it on to the system and set to the "running" status at the same time. The batch is initially created in a quantity of 1.000.000.000.

Output batch change

Output material may be changed for a running OP using the "output batch change" button.

The input batches that are logged on are displayed with their available remaining quantity in a list within the output batch change dialog. The following data may be entered:

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here. The selection refers to transport units that were assigned to the material type of the OP.

Comment on batch

In this field a comment may be saved for the batch to be produced.

Quantity

The batch to be produced is posted with the quantity entered here. The quantity is taken over as primary quantity of entry to the order and machine and a goods movement is generated as goods receipt.

Quality

A batch may be classified as yield or scrap quantity. The system posts yield batches with the "free" status. Scrap batches automatically get the "blocked" status. When scrap is selected, a valid scrap reason has to be assigned.

"Preceding batches" function key:

This button opens a list with output batches which have already been produced for this OP.

"Change inp. batch" function key:

Using this button the user can switch to the "input batch change" function.

New batch

When a current output batch is completed, a new batch number is simultaneously created for the next batch. The batch number may be assigned automatically or manually. The batch generated in this way is created with the batch number in the system and set to the "running" status.

Entry of additional batch attributes

Several additional batch attributes may be recorded for a material type by the configuration MPL --> Master data --> Attributes.

In case attributes are defined to be recorded at the terminal for the material type of the running OP, another input dialog is opened, when the output batch is changed. The dialog opens additionally after clicking OK in the output batch change function, interrupt OP and finish OP function.

Example when two additional attributes are collected:

Using batch attributes, numeric and alphanumeric values may be recorded which are then saved for the produced batch in an additional table.

Enter goods receipt batch

A new goods receipt batch may be created in the system by the "enter GR batch" button.

Having clicked the "OK" button, the batch is created and the dialog remains open for further entries.

Batch numbers may be generated automatically or manually depending on the configuration. Moreover, the following data are saved at the batch.

Workplace

Machine where the batch was recorded

Operation

Order where the batch was recorded

Material

Material number of the batch

Quantity and unit

The batch is created with the quantity and unit entered here. A goods receipt is posted with this quantity.

Quality

A batch may be classified as yield or scrap quantity. The system creates yield batches with the "free" status. Scrap batches automatically get the "blocked" status and the batch class "scrap". When scrap is selected, a valid scrap reason relating to the corresponding workplace has to be assigned.

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here.

Comment on batch

A comment may be saved for the goods receipt batch in this field.

Repost batch

Using the "repost batch" button, an existing batch may be reposted to another material buffer.

Having clicked the "OK" button, the batch is rebooked to a new material buffer and the dialog remains open for further entries.

As an alternative, the batch can also be reposted from yield to scrap.

4.4 Throughput batch mode

A HYDRA machine may also be configured in "throughput batch mode". In throughput batch mode input material is continued being processed with unchanged number (throughput batch number) via an OP.

The entry functions for throughput batch processing at the terminal are identical to those for active batch tracing at the machine.

Please note/restriction: At machines with “throughput batch mode” it is impossible to log operations on at the same time.

4.5 Manual partial upload in HYDRA-MPL environment

Using the manual partial upload function of HYDRA-ADE (A_TR dialog), it is possible to record a partial quantity for the current operation and thus for the active output batch.

The following performance as regards the entry of quantities results from a manual partial upload/confirmation:

- Scrap is only transferred to the active operation and not to the output batch
- Yield is accumulated on the current output batch
- In this case, log records of the record type “H“ are also generated when the shift changes, provided that yield has been recorded beforehand.
- In case yield has already been recorded as partial quantity for an output batch, it is no longer possible to log the batch off as scrap, when the output batch is changed the next time. In this case, the batch assigned to the “yield” class has first to be completed, before scrap can be posted.

Effects on available retrograde material components are as follows:

- When a total quantity is recorded, scrap is not deducted in a retrograde manner
- When yield/scrap is recorded, withdrawal also takes place when it comes to scrap
- Negative values, which might result from quantity offsetting, are not considered

The following notes and restrictions are to be taken into account:

- Personal partial uploads (with P_AB) are not taken into account
- Scrap quantities are only uploaded to a PPS system, provided that the interface has been configured for the “upload of partial confirmations”
- At the moment it is only supported to post quantities of the “yield” and “scrap” accounts.

- The partial upload itself does not trigger a goods movement. This is only the case, when the output batch has been completed.

- The checks relating to the input quantity and the display within the consumption balance cannot be used together with the "collect input quantity in relation to batches" configuration within the material type, as the current output batch is not completed for partial quantities. This affects, among other things, the collection of serial numbers with batches (MPL-SNR).

4.6 Display of produced output batches

Using the third list, it is possible to display the output batches produced for a running operation that is subject to batch management within the machine master at the terminal.

A maximum number of 20 output batches is displayed for each machine in the list. Yield and scrap batches are shown, which have been produced at this terminal since output batches were changed.

As this list is only kept locally by the respective terminal, it is not synchronized with the server, when AIP is started.

The list includes, among others, the article, article designation, batch number, date, time, quantity, batch class, user fields and alternative batch numbers. Individual fields might not be assigned values, which depends on the input scenario when generating output batches.

Configuration:

List contents can be configured via the [MNR_AMAT.LST] section in the ctaiplay.ini file.

Example:

CTAIPLAY.INI

```
[ MNR_AMAT.LST ]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=ROW.IDX=-
GRID_CAPTION=produced output batches

EXAMINE_SCANEXPR1=KLASSE=G
EXAMINE_SCANCOLOR1=clGreen
EXAMINE_SCANEXPR2=KLASSE=A
EXAMINE_SCANCOLOR2=clRed

; ROW.IDX=N10,50,R,Row
```

```
CNR=C20,150,L,batch number
;CNR=*CNR,batch
ATK=C25,125,L,article
; ATK=*ATK
; KLASSE=C3,40,Z,*
MENGE=N12.0,70,R,quantity
EINH=C3,30,Z,ME
DAT=dd.mm.yyyy,70,L,date
ZEI=hh:mm:ss,60,L,time
ATKBEZ=C30,200,L,article designation
```

<u>Display as third list:</u>

Figure: Display of produced output batches

Using the  button, the user can switch to the display of output batches.

Server-based comparison of produced output batches

As of the AIP program version V# 2.0.2.54 the list of produced output batches of a machine can be synchronized with the server using a cyclic comparison function.

The server comparison [Technically: PDM list 13 with MOD=P] is enabled by the configuration within the customized configuration file ”ctlisten.ini“.

Output batches recorded within the last three days at this terminal are only compared.

4.7 Batch information

Batch information is displayed in a dialog, when the “batch info” button is clicked.

4.8 Display of consumption balance

When logging an OP off, the “consumption balance” (V_BLZ dialog) may be displayed, which has to be configured via the machine and the material type of the operation.

The consumption balance is shown, provided that this option has been activated at the machine and material type.

The consumption balance shows the material consumption based on batches and the user is able to log running batches off.

Figure: Consumption balance – V_BLZ

Configuration:

A special configuration (function = “DLG=V_BLZ;BREAK-ON-CANCEL“) has to be defined for the OK button of the logoff dialog (e.g. A_AB_MPL) to make sure that the consumption balance is started, when the OP is logged off.

Sample configuration of the OK button for starting the consumption balance:

Figure: Example for the OK button configuration including consumption balance

Display of consumption postings

The “show details” function allows for the consumption quantities, which have been collected so far, to be displayed.

Figure: Display of consumption quantities – V_BLZ_DTL

Logging input batches off

By way of the “log batch off” function, the user can choose a currently running batch from the list and log it off by entering the consumption.

Figure: Log batch off – V_BLZ_CEAB

4.9 Advance logon of input batches

The AIP terminal provides the following dialogs or functions to log input batches on in advance. The function “advance logon of input batches” has been designed to be able to “set up” and log on the next input batch, while another OP and input batch are currently running. This next input batch is not yet active but assigned the flag “logged on in advance”.

An input batch can be logged on in advance for a currently running OP or for a prepared OP.

General

Usage

The process might require an input batch to be logged on in advance and set up accordingly on a machine, while the preceding input batch is still being used for a material.

This situation frequently occurs at very large machines processing, for example, roles or belts that are uncoiled as input batch at the beginning of the machine and coiled up as output batch at the end of the machine.

As the users are mostly busy with activities at the end of the machine at the time when the input batch actually needs to be changed, they cannot perform the input batch change and, as a result, they are provided with the opportunity to log the next input batch on already in advance for an order/OP.

Then the input batch can actually be changed by logging a new OP on or a project-specific call can be established.

Configuration

The settings required for using the function “advance logon of input batches” is described here.

Process/procedure

The procedure for using the function “advance logon of input batches“ or the logical process is described here.

AIP usage

AIP basic screen

The basic AIP screen shows the function key “Advance logon of input batch” (preregistration of input batch). The dialog for logging input batches on in advance may be used by clicking this function key.

Advance logon of input batches (CE_VWL_MPL)

The user selects the workplace to which an input batch is to be logged on in advance in the basic screen. The below dialog (CE_VWL_MPL) opens by clicking the function key ”Advance logon of input batch” (preregistration of input batch).

If an operation is currently running/logged on to the workplace, this one will be selected by default. The input batch (that is to be logged on in advance) is entered/scanned for the selected BOM item. Advance logon of input batches is started by clicking the button “post batch”.

At first the input batch is checked for validity (dialog CE_VAN). The material number of the input batch is checked against the material number of the component list or the BOM item. The input batch is logged on in advance, once the button “log input batch on in advance” has been clicked.

Finally, the input batch that has been logged on in advance is displayed in purple in the BOM of the component. The dialog is exited by the “cancel” key**.**

Show third list

Besides the logged on/running input batches, the third list of the basic screen also shows the input batch that has been logged on in advance and highlights it in purple.

Consequently, a BOM item can use a logged on/running input batch and an input batch logged on in advance at the same time.

Log off batch logged on in advance

The dialog “advance logon of input batches“ also provides the opportunity to log off or reset an input batch that has been logged on in advance.

To do so, the input batch that has been logged on in advance is selected once more and the user confirms the button “post batch“.

The input batch is logged off by the dialog CE_VAB.

Log on input batch logged on in advance with OP

When logging a prepared operation on, an input batch that already has been logged on in advance is shown for a BOM item. The input batch that has been logged on in advance is logged on automatically by logging the operation on.

If, however, another input batch is logged on for this BOM item, the input batch logged on in advance is kept for the running OP.

When an OP is logged off/interrupted, all input batches logged on in advance are also logged off/reset automatically.

4.10 Recording of serial numbers

Serial numbers are assigned to be able to distinguish between individual items of material. Consequently, the combination of material number and serial number uniquely identifies an individual item.

MES provides several variants including different features to record serial numbers:

- Variant 1: Entry of serial numbers for OPs that are not subject to batch management (dialog A_SNR/“E“ only)
- Variant 2: Entry of serial numbers for OPs that are subject to batch management (dialog A_SNR)
    - Manual input of the serial number. It is assigned to a HYDRA batch number.
    - Automatic assignment of the serial number = HYDRA batch number
    - Automatic assignment of the serial number. It is assigned to a HYDRA batch number

4.10.1 Entry of serial number for OPs that are not subject to batch management (dialog A_SNR/ “E“)

General/utilization

Serial numbers are assigned in ERP when creating a production order. The assigned serial numbers are transferred as detailed information for the order header where they are managed accordingly in the interface to HYDRA.

The OP of the order that is not subject to batch management is logged on and the relevant serial numbers are recorded with the relevant quality (yield/scrap).

The serial number is entered manually (e.g. also barcode for the individual item) by the user on AIP. To simplify data collection, the serial numbers that are still available for the order are shown in a list where they can be selected. Please note that a serial number can occur as scrap only once.

The quantity recorded for the order/OP is always 1 for each individual item and is posted onto the relevant yield or scrap account of the order/OP subject to the quality.

Uploads (goods movements) relating to serial numbers can be transferred to the ERP system for recorded serial numbers/individual items.

Configuration

The required settings for using the function “collection of serial numbers” (without orders that are subject to management in batches) are described here.

Procedure

The procedure for using the function “collection of serial numbers” (without orders that are subject to management in batches) or the logical process are described here.

Utilization with AIP

Basic AIP screen

The operator uses the dialog “serial numbers” (A_SNR). The relevant function key is configured in the basic view of AIP.

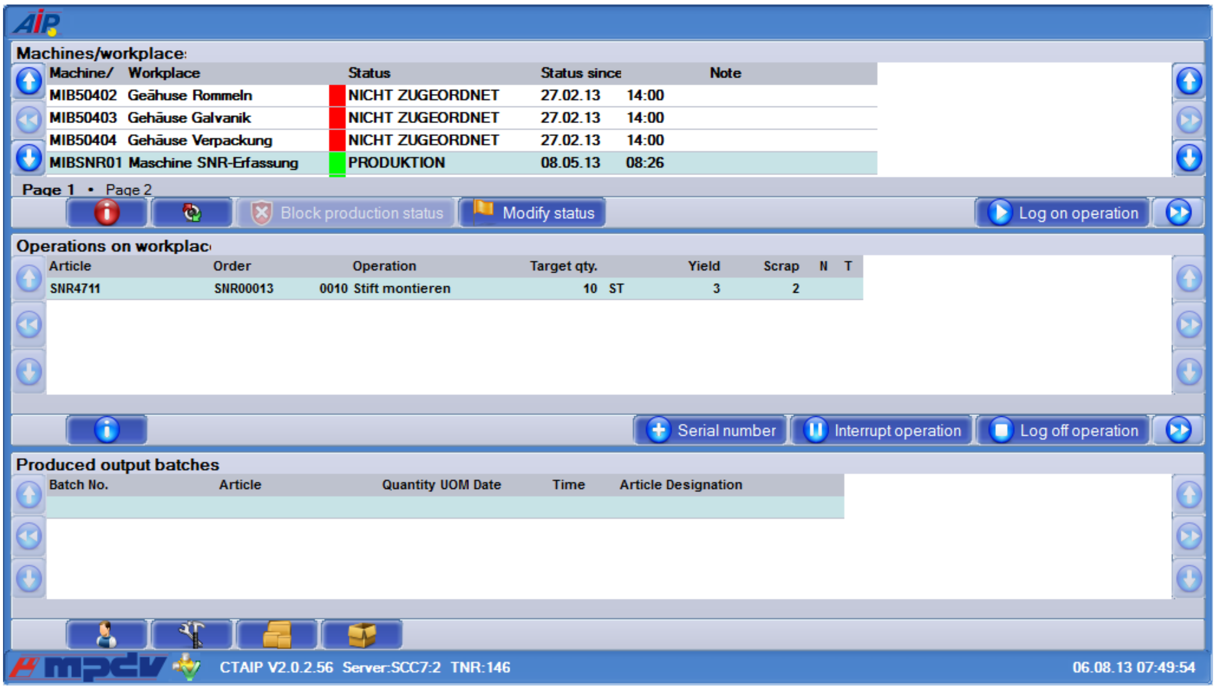


Figure: Basic view including function key “serial number“ (A_SNR)

Entry of serial numbers (A_SNR)

Figure: Entry of serial numbers on the terminal (A_SNR/ “E“)

Serial number

The user enters the serial number. The serial number has to be part of the order. A validation check takes place. If the serial number is not part of the order or if it already has a result (already recorded) it cannot be used.

By clicking , a selection list showing the “free“ serial numbers for the order is displayed.

Requesting the selection list simultaneously updates the list of serial numbers that have already been recorded.

Grid

The displayed list shows all serial numbers that have already been entered including their relevant quality (yield/scrap).

- Scrap is shown red
- Yield is shown green

Quality

If a serial number is entered, it has to be assigned quality. The following can be chosen:

- Yield
  The serial number is entered with the quality “yield”. The operation quantity is increased by yield = 1. The default value is “yield”.

- Scrap
  The serial number is entered with the quality “scrap”. The operation quantity is increased by scrap quantity = 1.

- Scrap reason
  If the quality is entered as scrap the user also has to enter or select a scrap reason.

“Capture“ function

By clicking this function, the entered serial number is recorded with the selected quality.

“List“ function

By clicking this function, the entire list of serial numbers that have already been entered including the selected quality can be updated (e.g. if the dialog is opened anew).

Interrupt/finish OP for orders with serial numbers (A_UN/ A_AB)

Quantities are not entered when operations subject to management in serial numbers are logged off. The quantity fields have to be disabled in these dialogs.

4.10.2 Entry of serial numbers for OPs subject to management in batches (dialog A_SNR)

General/utilization

If an order/operation subject to management in batches is used to enter serial numbers, there are the following alternatives to enter or assign the serial numbers for data collection:

- Manual input of the serial number. It is assigned to a HYDRA batch number.
- Automatic assignment of the serial number = HYDRA batch number
- Automatic assignment of the serial number. It is assigned a HYDRA batch number

Configuration

The required settings to use the function “collection of serial numbers” (without orders that are subject to management in batches) are described here.

Procedure

The procedure to be able to use the function “collection of serial numbers” (without orders that are subject to management in batches) or the logical process are described here.

AIP usage

Entry of serial numbers (A_SNR) including manual assignment of serial numbers – characteristic “E“

If batch management requirement is enabled for the operation, a batch is created additionally for each serial number in the material and production logistic.

Consequently, the relation between the current output batch and the serial number posted in the database are saved for tracing. In this case, the output batch is considered as ID without quantity and, as a result, it does not have a quantity.

Batch attributes that might be required to be recorded for OPs handled in batches are recorded by way of the commonly used standard dialog before sending.

The “collection of serial numbers“ dialog is structured as described below. The serial number can be entered. Batch assignment is performed in the background.

Entry of serial numbers (A_SNR) including automatic assignment of serial numbers – characteristic “G“

By using the option “serial number requirement” = “G” for the operation, output batches are entered as serial number. In this case, the serial number is the output batch number.

The field “serial number“ is disabled and includes the current output batch. Posting is possible, once the badge number has been entered by the function “capture”. An applicable scrap reason has to be entered, provided that a “quality indicator” unequal to “yield” has been chosen.

Exactly one batch with quantity 1 is created for each serial number and the serial number matches the batch number in this case.

The function “list“ updates the list of serial numbers that have already been recorded.

Batch attributes that might be required to be entered for OPs handled in batches are recorded by the common standard dialog before sending.

Entry of serial numbers (A_SNR) including automatic assignment of serial numbers by the number range - characteristic - ”S“

Serial numbers in relation to a batch are recorded by the option “serial number requirement” = “S” at the operation. For this purpose, the operation needs to be subject to batch management. An output batch change (CA_WL) is triggered every time a serial number is posted. This output batch change creates a batch with quantity 1 for the serial number in MPL.

By clicking the function  a new serial number is determined on the server and shown in the field “serial number”. The new serial number is assigned uniquely for the entire system by the number range SNR.

Batch attributes that might be required to be entered for OPs handled in batches are recorded by using the standard dialog before sending.

Interrupt/finish operation for orders including serial number tracking

Quantities are not entered when operations subject to management in serial numbers are logged off. The quantity fields are to be disabled in these dialogs.

5 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
--------	-------------------	---- General ---
00.	00.ABC123	Data not defined, ➔ 00. will be deleted and data “ABC123” will be passed to standard processing
01.	01.OK 01.ESC	Action barcode ➔ Dialog cancelled or ended with OK button or Esc button.
--------	-------------------	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Suborder number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16.200	Machine → Acronym <MNR> ➔ Passed to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> ➔ Passed to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> ➔ Passed to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > ➔ Passed to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> ➔ Passed to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
--------	-------------------	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40.100 40.MONTAGE	Destination → Acronym <ZLO> ➔ Passed to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> ➔ Passed to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→➔
43.		Throughput batch number → Acronym <DLL>→➔
44.		Alternative batch number → Acronym <CNR:ALT1>→➔
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
--------	-------------------	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> ➔ Passed to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> ➔ Passed to dialog with FGR=1 or FGR=1001
--------	**-------------------**	**---- Customer-specific barcodes ---**
90.		
…		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]
BarcodePrefixChar=$

If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

5.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR **BarKenn91=EGR:GUT**	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345“ → SAPCNR=12345

Firmly assigned barcode prefixes:
10:ANR
11:AUNR
12:AFOLG
13:AGNR
14:UAGNR
15:RMNR
16:MNR

17:MST
18:AUSGRD
19:AGGGRD
20:BPOS
21:LPKZ
22:SPLNR
40:ZLO
41:TPE
42:CNR
43:DLL
44:CNR:ALT1
45:CNR:ALT2
46:CNR:ALT3
47:CNR:ALT4
48:CNR:ALT5
49:CNR:ALT6
50:KNR
51:PNR
52:KST
53:FGR

6 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.

Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

6.1 Basic configuration

Entry	Comment
Section [system]	

Entry	Comment
Section [system]	
`Usr=21`	Distinct terminal number

`Hypath=d:\hydra\`
 	
`Hypath=/usr/hydra/`
	HYDRA server path:
	
	Windows NT:
	-> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed
	Unix:
	->Unix notation

`Hostname=192.9.200.24`	Internet address of the server
`Offlinetimeout=600`	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
`Showcursor=on`	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
`Loadfile=` `ctnet\win\ctaip.txt`	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within "hypath")
`Watchdog=on`	ON: Watchdog is activated OFF: Watchdog is not activated
`Demo=off`	'on': Offline demo mode; always off in the production environment
`parameters=-t`	The –t parameter switches off the virtual keyboard
`TMOUT_C=xxx`	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_S=xxx`	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_R=xxx`	Timeout for RECEIVE of the server If not specified, default = 120 seconds

Entry	Comment
`TMOUT_F=xxx`	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	**Configuration of customized barcode prefixes.**

`BarKenn90=MNR4`
`...`
`BarKenn99=ANR3`
	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.

Section [comports]	

`Com1=0`
`com2=MSS`
`com3=BAR`
`Com3=LEGIC`
`Com4=RFLESER`
	Assignment of serial interfaces to the connected devices
	MSS – machine interface
	BAR, LEGIC, RFLESER – various reading devices

Section [MSS-INIT]	
`ZAEHLER=|1|2|3|4|5|6|7|8|`	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER=|1|2|3|4| IN=|5|6|7|8|
`IN=|9|10|11|12|13|14|15|`	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
`CHARGE=|5|6|`	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
`MSSZyklusBerechnung=ON` `MSSZyklusReferenz=0`	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
`CalculateCycle=ON`	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.
`WochenEnde_ProdCheck=ON`	This function prevents the weekend automatic from affecting the “production“ status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.

Entry	Comment
sFrom999ToNotAttributed=OFF	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the “no monitoring” option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
Modus=PDV Modus=PDV,BDE	Operation as interface IOP←→DOS terminal (standard) Operation as shop floor terminal with PDV
Terminals=121,122,123	CTDOS terminals “connected” by LAN. Only required with Mode=PDV
PDVTerminalDir=c:\hsrv\spool\ PDVIOPDir=c:\IOPSim\	Directory for communication with CTDOS terminals Directory for communication with the IOP
For debug purposes only: InfoFenster=100 SlowDown=600	 Number of lines of the “current” window (presentation of last PDV actions) Slow down, to make events “visible”
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.

Entry	Comment
`BarcodeNest=`	This field specifies which acronym is entered into the cavity number field by the scanner.
`BarcodeNumm=`	This field specifies which acronym is entered into the number field by the scanner.

7 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version
[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section [section <TNR-USER>]
<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User

The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
`FollowExternStatus=on`	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
`InstallFonts=on`	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
`OnlyInstallFontsAfterDownload=false`	"`InstallFonts=on`": If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
`InstallTvicport=on`	If "off" the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

`AttachedApplication=First`	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: **First** → search for connected application first **AfterUserViewer** → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) **Last** → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (**default**). **Off** → Connected application is never searched.
`HTTPBrowser=standard`	Viewing of documents (via OP info): If documents are configured with a path of the type "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
`SupressErrorMessage=70012`	Suppress message "material is not planned"
[SignatureRecording->User 0]	
`ManualBadgeInput=true`	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) **true** → activates keyboard input for the "user" field in the terminal
`Transparency=255`	The signature dialog can also be transparent. **`255`** → Signature dialog is 0 % transparent (not transparent) **`1`** → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)

`ShowPosition=TR`	The position of the signature dialog can be adjusted as follows: **TL** Top – Left **TM** Top – Middle **TR** Top – Right **ML** Middle – Left **MM** Middle – Middle (Default) **MR** Middle – Right **BL** Bottom – Left **BM** Bottom – Middle **BR** Bottom – Right Available as of CTAIP (V# 2.0.2.25)
`USE_SERVICE_ACCOUNT=1`	**0** (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO). **Please note:** ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
`SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY`	**`REPORTING_USER_READONLY`** The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. **`REPORTING_USER_CHANGEABLE`** The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

SIGNATURE_1_LOGON_TYPE=HYDRA	**"" / Not set / "EMPTY"** There is also an alternative login procedure. **HYDRA** The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. **ACTIVEDIRECTORY** The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. **MIXED_BUT_UNIQUE** Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO“ enabled → Windows only "SSO“ disabled → HYDRA only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures with the terminal in the area of quality data collection.

7.1 Layout configuration

Entry	Comment
Section and/or [terminal configuration 0] [terminal configuration 2XXX];	**(general configuration) (2XXX terminal-specific configuration)**
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON	As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)

Entry	Comment
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;**(general configuration)** [QRD-PRINTER->TICKET 2xxx] ;**(2XXX configuration for a specific terminal)**	
COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured **<Events>,** if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >	Reloads the order row for the configured **<Events>,** if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
COMPLETE-..-EVENT=< Events > **COMPLETE-..-EVENT=#ALL#** **COMPLETE-..-EVENT=A_AN|A_P_AN**	Explanation on the configuration of **<Events>** ➔ Using **<#ALL#>** the row (ANR/MNR) that is not available is reloaded for any event. ➔ **<A_AN|A_P_AN>** restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.